The following citati

User

The following citations were found in the files provided by the user:

Citation 1: "[] **[1:15168:13] INDICATOR-COMPROMISE Suspicious .ru dns query** []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.002972 192.168.1.96:49996 -> 192.168.1.1:53
UDP TTL:128 TOS:0x0 ID:3338 IpLen:20 DgmLen:54
Len: 26

[] **[1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection** []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:11.994886 192.168.1.96:49322 -> 104.27.158.125:80
TCP TTL:53 TOS:0x0 ID:23760 IpLen:20 DgmLen:1994 DF
A*** Seq: 0xCBEB269 Ack: 0x3960D09 Win: 0x8800 TcpLen: 20

[] **[1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection** []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:11.994487 192.168.1.96:49318 -> 104.20.221.29:80
TCP TTL:51 TOS:0x8 ID:44202 IpLen:20 DgmLen:1289 DF
A*** Seq: 0xEC586A97 Ack: 0x717262C7 Win: 0x8000 TcpLen: 20"

Citation 2: "[] **[139:1:1] (spp_sdf) SDF Combination Alert** []
[Classification: Senstive Data] [Priority: 2]
06/27-13:44:12.816109 202.93.17.181 -> 192.168.1.96
PROTO:254 TTL:128 TOS:0x0 ID:4389 IpLen:20 DgmLen:20 DF

[] **[1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection** []
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06/27-13:44:12.785996 192.168.1.96:49500 -> 104.20.166.37:80
TCP TTL:51 TOS:0x8 ID:65198 IpLen:20 DgmLen:1815 DF
A*** Seq: 0x556539F9 Ack: 0xCE719B6A Win: 0x8400 TcpLen: 20

[] **[1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection** []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.657872 192.168.1.96:49475 -> 95.213.255.23:80
TCP TTL:39 TOS:0x8 ID:38181 IpLen:20 DgmLen:1293 DF
A*** Seq: 0x567F5BB5 Ack: 0x354677E6 Win: 0x8000 TcpLen: 20"

Citation 3: "[] **[1:15168:13] INDICATOR-COMPROMISE Suspicious .ru dns query** []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.048047 192.168.1.96:61354 -> 192.168.1.1:53
UDP TTL:128 TOS:0x0 ID:3419 IpLen:20 DgmLen:54
Len: 26

[] **[1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection** []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.020744 192.168.1.96:49339 -> 104.20.151.6:80
TCP TTL:51 TOS:0x8 ID:39403 IpLen:20 DgmLen:817 DF
A* Seq: 0xD1C1E240 Ack: 0x41FEEA3E Win: 0x7C00 TcpLen: 20

[] **[1:23832:4] INDICATOR-OBFUSCATION non-alphanumeric javascript detected** []
[Classification: Attempted User Privilege Gain] [Priority: 1]
06/27-13:44:12.060130 104.20.151.6:80 -> 192.168.1.96:49339
TCP TTL:51 TOS:0x8 ID:39404 IpLen:20 DgmLen:4916 DF
AP Seq: 0x41FEEA3E Ack: 0xD1C1E549 Win: 0x1F TcpLen: 20
[Xref => http://patriciopalladino.com/blog/2012/08/09/non-alphanumeric-javascript.html]"

Use the citations above to respond to the user query, only if they are relevant. Otherwise, respond to the best of your ability without them.

User Query:

Your role is SOC intrusion analyst. Review the attached CW1Snort.txt and CW1Suricata.txt logs. Identify the compromised internal IP address, group the high priority alerts to understand what the infection is, and give me the exact Ubuntu Linux grep commands and Wireshark display filters needed to curate reproducible packet evidence for an incident report. Write concisely and holistically.

- Attachment: file:///C:/Users/dtoader/.lmstudio/user-files/1794309513921%20-%20585.txt (227.7 KB)
- Attachment: file:///C:/Users/dtoader/.lmstudio/user-files/1794309513922%20-%20620.txt (295.3 KB)

Assistant

Compromised Internal IP

`192.168.1.96` (Source of all Priority 1 outbound alerts and DNS indicators)

Infection Analysis & Alert Grouping

CNC `Win.Trojan.Pushdo` variant. All high-priority alerts originate from this host, forming a coherent TTP sequence:

1. **Domain Compromise/Reconnaissance:** Suspicious `.ru` DNS queries to internal resolver (`192.168.1.1`) indicate initial beaconing or malicious domain resolution.
2. **Web-Delivered Infection & Obfuscation:** Non-alphanumeric JavaScript detection in HTTP responses points to a browser/web-proxy delivery mechanism with payload obfuscation.
3. **C2 Communication:** Repeated outbound TCP/HTTP callbacks to multiple external IPs on port 80 (`104.27.x`, `104.20.x`) confirm active command-and-control communication, likely using HTTP steganography or web tunneling.
4. **Secondary Indicator:** Priority 2 SDF alert on port 254 from `202.93.17.181` suggests additional data exfiltration or a secondary sinkhole being probed.

The alerts group cleanly into one infection lifecycle: **Delivery → Obfuscated Beaconing → C2 Callbacks**.

Exact Ubuntu Linux Grep Commands

Run these in the SIEM log directory to isolate high-priority, host-specific malware indicators and preserve temporal context:

```
# Extract all Priority 1 malware & indicator alerts for the compromised host
grep -iE "(MALWARE-CNC|INDICATOR-COMPROMISE|INDICATOR-OBFUSCATION)" CW1Snort.txt | grep "192.168.1.96"

# Suricata-specific extraction (identical pattern matching)
grep -iE "(MALWARE-CNC|INDICATOR-COMPROMISE|INDICATOR-OBFUSCATION)" CW1Suricata.txt | grep
"192.168.1.96"

# Isolate the obfuscated HTTP response flow for malware reverse engineering
grep -A5 "INDICATOR-OBFUSCATION" CW1Suricata.txt | grep "104.20.151.6"
```

Wireshark Display Filters

Apply these to reproduce and triage the packet evidence efficiently:

```
# Primary C2 & beaconing traffic (HTTP port 80)
tcp.port == 80 and host == 192.168.1.96

# DNS anomaly / .ru resolution detection
udp.port == 53 and ip.hostname == ".ru" or ip.addr == 192.168.1.96 and udp.port == 53

# Holistic capture (recommended for incident report baselining)
host == 192.168.1.96 and (tcp.port == 80 || udp.port == 53)
```

Evidence Curation Tip: After applying the holistic Wireshark filter, extract HTTP request bodies to verify obfuscation using:
`tsh -i <capture> "display.http.request" | grep -i "javascript\|non-alphanumeric"`

This isolates the payload used for initial infection and confirms web-delivered trojan behavior before malware reverse engineering.

User

The following citations were found in the files provided by the user:

Citation 1: "**[] [139:1:1] (spp_sdf) SDF Combination Alert []**
[Classification: Senstive Data] [Priority: 2]
06/27-13:38:34.536182 119.28.70.207 -> 192.168.1.96
PROTO:254 TTL:128 TOS:0x0 ID:661 IpLen:20 DgmLen:20 DF

[] [1:15306:22] FILE-EXECUTABLE Portable Executable binary file magic detected []
[Classification: Potential Corporate Privacy Violation] [Priority: 1]
06/27-13:43:52.407982 145.131.10.21:80 -> 192.168.1.96:49190
TCP TTL:239 TOS:0x20 ID:41885 IpLen:20 DgmLen:1207
AP Seq: 0xA90C366B Ack: 0xCB5AB7FC Win: 0x2180 TcpLen: 20

[] [1:15306:22] FILE-EXECUTABLE Portable Executable binary file magic detected []
[Classification: Potential Corporate Privacy Violation] [Priority: 1]
06/27-13:43:52.407982 145.131.10.21:80 -> 192.168.1.96:49190
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Citation 2: "**[] [139:1:1] (spp_sdf) SDF Combination Alert []**
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06/27-13:44:12.816109 202.93.17.181 -> 192.168.1.96
PROTO:254 TTL:128 TOS:0x0 ID:4389 IpLen:20 DgmLen:20 DF

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[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.785996 192.168.1.96:49500 -> 104.20.166.37:80
TCP TTL:51 TOS:0x8 ID:65198 IpLen:20 DgmLen:1815 DF
A* Seq: 0x556539F9 Ack: 0xCE719B6A Win: 0x8400 TcpLen: 20

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06/27-13:44:12.657872 192.168.1.96:49475 -> 95.213.255.23:80
TCP TTL:39 TOS:0x8 ID:38181 IpLen:20 DgmLen:1293 DF
A* Seq: 0x567F5BB5 Ack: 0x354677E6 Win: 0x8000 TcpLen: 20"

Citation 3: "**[] [139:1:1] (spp_sdf) SDF Combination Alert []**
[Classification: Senstive Data] [Priority: 2]

06/27-13:43:59.242880 59.106.164.230 -> 192.168.1.96
PROTO:254 TTL:128 TOS:0x0 ID:1061 IpLen:20 DgmLen:20 DF

[] [120:8:2] (http_inspect) INVALID CONTENT-LENGTH OR CHUNK SIZE []
[Classification: Unknown Traffic] [Priority: 3]
06/27-13:44:01.462986 192.168.1.96:49200 -> 96.82.200.1:80
TCP TTL:128 TOS:0x0 ID:1284 IpLen:20 DgmLen:40 DF
A* Seq: 0x763F0FC7 Ack: 0xC5FE1D52 Win: 0xFB6D TcpLen: 20
[Xref => http://cve.mitre.org/cgi-bin/cvename.cgi?name=2013-2028]

[] [1:28629:6] INDICATOR-OBFUSCATION obfuscated script encoding detected []
[Classification: Misc activity] [Priority: 3]
06/27-13:44:01.624079 148.251.33.194:80 -> 192.168.1.96:49194
TCP TTL:43 TOS:0x0 ID:37796 IpLen:20 DgmLen:11720 DF
A* Seq: 0xA2F1C810 Ack: 0x95857EE1 Win: 0x7F TcpLen: 20"

Use the citations above to respond to the user query, only if they are relevant. Otherwise, respond to the best of your ability without them.

User Query:

What is C2? What is SDF? What is -iE in grep? Should grep use the IP address if that's what I'm trying to find out?

Assistant

Based on the provided citations and standard cybersecurity terminology, here are the concise answers to your questions:

1. What is C2?

C2 (Command and Control) refers to a communication channel used by malware or attackers to maintain control over compromised systems. In the logs above, the high-priority alerts show outbound TCP connections from `192.168.1.96` to external IPs on port 80 (e.g., `104.20.x`, `95.213.x`). This indicates the malware is beaconing or receiving instructions via HTTP, a common technique for **HTTP C2** or web tunneling.

2. What is SDF?

SDF refers to **Snort Detection Framework**. It is an open-source intrusion detection system (IDS) developed by Snort. The tag `(spp_sdf)` in the citations indicates that these alerts were generated by a rule set within the Snort framework, specifically related to **Safe Data Filtering** or specific combination rules used to correlate different indicators of compromise.

3. What is `-iE` in grep?

In the command `grep -iE ...`:

- **-i**: Case-insensitive matching (e.g., matches "malware" and "MALWARE" equally).
- **-E**: Extended regular expressions (allows for more complex patterns using parentheses and alternation).

4. Should grep use the IP address if that's what I'm trying to find out?

Yes. If your goal is to isolate traffic related to a specific host, **filtering by its IP address is essential**. In these citations, `192.168.1.96` is consistently identified as the compromised internal host (source of malware outbound connections and indicator alerts). Filtering by this IP ensures you are analyzing traffic relevant to the infection source rather than unrelated network noise.

Example:

```
grep -iE "MALWARE-CNC|INDICATOR" CW1Snort.txt | grep "192.168.1.96"
```

User

Important: No citations were found in the user files for the user query. In less than one sentence, inform the user of this. Then respond to the query to the best of your ability.

User Query:

Teach me to write a grep command to capture just the line that has the 'Priority' text

Assistant

No citations were found in the provided files for the user query.

Use `grep -i "priority"` to extract lines containing "Priority" (case-insensitive), or `grep "^.*[Pp]riority.*$"` if you want to ensure the word appears as a standalone term within the line.

User

Important: No citations were found in the user files for the user query. In less than one sentence, inform the user of this. Then respond to the query to the best of your ability.

User Query:

What are other grep options?

Assistant

No citations were found in the provided files for the user query.

Other useful `grep` options include:

- `-n` : Print line numbers (`grep -n "pattern" file`).
- `-c` : Count matching lines (`grep -c "pattern" file`).
- `-v` : Invert match (show non-matching lines) (`grep -v "pattern" file`).
- `-p` : Print only matched portions, not the entire line.
- `-A N` / `-B N` : Show N lines after/before a match.
- `-E` : Extended regex syntax (as mentioned earlier).

User

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PROTO:254 TTL:128 TOS:0x0 ID:4389 IpLen:20 DgmLen:20 DF

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[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.785996 192.168.1.96:49500 -> 104.20.166.37:80

TCP TTL:51 TOS:0x8 ID:65198 IpLen:20 DgmLen:1815 DF
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TCP TTL:39 TOS:0x8 ID:38181 IpLen:20 DgmLen:1293 DF
A* Seq: 0x567F5BB5 Ack: 0x354677E6 Win: 0x8000 TcpLen: 20"

Citation 3: "**[] [1:11192:20] FILE-EXECUTABLE download of executable content []**
[Classification: Potential Corporate Privacy Violation] [Priority: 1]
06/27-13:43:52.407756 145.131.10.21:80 -> 192.168.1.96:49190
TCP TTL:128 TOS:0x0 ID:835 IpLen:20 DgmLen:17427 DF
A* Seq: 0xA90C3547 Ack: 0xCB5AB7FC Win: 0xFD20 TcpLen: 20
[Xref =>
http://www.microsoft.com/smallbusiness/resources/technology/security/practice_safe_computing_and_thwart_online_thugs.mspx]

[] [139:1:1] (spp_sdf) SDF Combination Alert []
[Classification: Senstive Data] [Priority: 2]
06/27-13:43:53.185747 145.131.10.21 -> 192.168.1.96
PROTO:254 TTL:128 TOS:0x0 ID:868 IpLen:20 DgmLen:20 DF

[] [1:28406:1] MALWARE-CNC Win.Trojan.Kazy variant outbound connection []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:43:54.128138 192.168.1.96:49191 -> 143.95.151.192:80
TCP TTL:49 TOS:0x8 ID:1431 IpLen:20 DgmLen:188 DF
A* Seq: 0xA88DB23D Ack: 0x2FD3568D Win: 0x3E00 TcpLen: 20
[Xref =>
http://www.virustotal.com/en/file/a064a1d3d8b9d8ab649686b7fb01e0631e569412388084f5c391722c98660763/analysis/]"

Use the citations above to respond to the user query, only if they are relevant. Otherwise, respond to the best of your ability without them.

User Query:

Please provide a prioritised list of the most important alerts directly from the log files.

Assistant

No citations were found in the provided files for the user query.

Based on standard SOC triage practices and the structure of the provided log citations, here is a prioritized list of the most important alerts to investigate:

Priority 1 Alerts (Critical)

- **MALWARE-CNC Win.Trojan.Pushdo variant outbound connection**
  - **Why:** Confirmed network trojan activity with command-and-control (C2) callbacks. This indicates an active infection and potential data exfiltration or further lateral movement.
  - **Action:** Immediate containment, malware sample analysis, and C2 server blocking.
- **FILE-EXECUTABLE Portable Executable binary file magic detected / download of executable content**
  - **Why:** Indicates delivery of a suspicious PE file (potential dropper or payload). High risk of privilege escalation or persistence mechanisms.
  - **Action:** Forensic analysis of the downloaded file, check for scheduled tasks/startup items, and monitor for behavior changes.
- **INDICATOR-COMPROMISE Suspicious .ru dns query**
  - **Why:** `.ru` domains are frequently associated with Russian state-sponsored or criminal actor groups. Indicates initial access or infrastructure compromise.
  - **Action:** Block the domain, review DNS logs for other compromised hosts, and check for hidden persistence.

Priority 2 Alerts (High)

- **(spp_sdf) SDF Combination Alert [Classification: Senstive Data]**
  - **Why:** Indicates potential data breach or unauthorized access to sensitive information. Requires immediate verification of what data was accessed and by whom.
  - **Action:** Review audit logs, assess data exposure, and initiate incident response protocols for a possible security violation.

Priority 3 Alerts (Medium)

- **(http_inspect) INVALID CONTENT-LENGTH OR CHUNK SIZE**
  - **Why:** Suggests malformed HTTP traffic, which could be an attempt to evade detection, inject malicious content, or exploit vulnerabilities in web services.
  - **Action:** Inspect the request/response headers and payloads for injection attempts or anomalies.

Priority 4 Alerts (Low)

- **(http_inspect) obfuscated script encoding detected**
  - **Why:** Indicates potential use of JavaScript or other scripts to hide malicious code, often used in web-delivered malware.
  - **Action:** Monitor for execution attempts and analyze the encoded content if possible.

Recommended Grep Commands for Further Analysis

```
# Extract all high-priority malware and indicator alerts
grep -iE "(MALWARE-CNC|INDICATOR-|FILE-EXECUTABLE)" CW1Snort.txt | grep "Priority: 1"

# Isolate C2 communication traffic
grep -i "outbound connection" CW1Suricata.txt | grep "192.168.1.96"

# Review sensitive data access alerts
grep -i "Sensitive Data" CW1Snort.txt
```

User

The following citations were found in the files provided by the user:

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[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.048047 192.168.1.96:61354 -> 192.168.1.1:53
UDP TTL:128 TOS:0x0 ID:3419 IpLen:20 DgmLen:54
Len: 26

[] **[1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection** []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.020744 192.168.1.96:49339 -> 104.20.151.6:80
TCP TTL:51 TOS:0x8 ID:39403 IpLen:20 DgmLen:817 DF
A*** Seq: 0xD1C1E240 Ack: 0x41FEEA3E Win: 0x7C00 TcpLen: 20

[] **[1:23832:4] INDICATOR-OBFUSCATION non-alphanumeric javascript detected** []
[Classification: Attempted User Privilege Gain] [Priority: 1]
06/27-13:44:12.060130 104.20.151.6:80 -> 192.168.1.96:49339
TCP TTL:51 TOS:0x8 ID:39404 IpLen:20 DgmLen:4916 DF
AP Seq: 0x41FEEA3E Ack: 0xD1C1E549 Win: 0x1F TcpLen: 20
[Xref => http://patriciopalladino.com/blog/2012/08/09/non-alphanumeric-javascript.html]"

Citation 2: "[] **[1:15168:13] INDICATOR-COMPROMISE Suspicious .ru dns query** []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.002972 192.168.1.96:49996 -> 192.168.1.1:53
UDP TTL:128 TOS:0x0 ID:3338 IpLen:20 DgmLen:54
Len: 26

[] **[1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection** []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:11.994886 192.168.1.96:49322 -> 104.27.158.125:80
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A*** Seq: 0xCBEB269 Ack: 0x3960D09 Win: 0x8800 TcpLen: 20

[] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:11.994487 192.168.1.96:49318 -> 104.20.221.29:80
TCP TTL:51 TOS:0x8 ID:44202 IpLen:20 DgmLen:1289 DF
A* Seq: 0xEC586A97 Ack: 0x717262C7 Win: 0x8000 TcpLen: 20"

Citation 3: "**[] [1:23832:4] INDICATOR-OBFUSCATION non-alphanumeric javascript detected []**
[Classification: Attempted User Privilege Gain] [Priority: 1]
06/27-13:44:08.765489 104.27.139.76:80 -> 192.168.1.96:49292
TCP TTL:53 TOS:0x0 ID:15927 IpLen:20 DgmLen:5129 DF
AP Seq: 0xCECA6E07 Ack: 0x234866D1 Win: 0x1F TcpLen: 20
[Xref => http://patriciopalladino.com/blog/2012/08/09/non-alphanumeric-javascript.html]

[] [1:23832:4] INDICATOR-OBFUSCATION non-alphanumeric javascript detected []
[Classification: Attempted User Privilege Gain] [Priority: 1]
06/27-13:44:08.816660 104.27.139.76:80 -> 192.168.1.96:49294
TCP TTL:53 TOS:0x0 ID:37546 IpLen:20 DgmLen:4782 DF
AP Seq: 0xAC4969A0 Ack: 0x35DA69A4 Win: 0x1F TcpLen: 20
[Xref => http://patriciopalladino.com/blog/2012/08/09/non-alphanumeric-javascript.html]

[] [1:15168:13] INDICATOR-COMPROMISE Suspicious .ru dns query []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:11.994287 192.168.1.96:51688 -> 192.168.1.1:53
UDP TTL:128 TOS:0x0 ID:3326 IpLen:20 DgmLen:54
Len: 26"

Use the citations above to respond to the user query, only if they are relevant. Otherwise, respond to the best of your ability without them.

User Query:

'Suspicious .ru dns query'

Is this necessarily suspicious? If so, why?

Assistant

No, not necessarily. While `.ru` domains are sometimes associated with state-sponsored actors or criminal groups, they can also be legitimate domains owned by Russian entities or individuals. The suspicion arises because:

1. **Unusual Geographic Origin**: In the context of a cybersecurity incident, unexpected DNS queries to foreign TLDs (like `.ru`) from internal devices may indicate reconnaissance or beaconing activity.

2. **Potential Malware Indicators**: These domains are occasionally used by malware for C2 communication or file distribution, especially if they are newly registered or part of a known malicious infrastructure.

To determine if the DNS query is truly suspicious, further analysis should include:

- Checking domain reputation and historical data.
- Verifying if the internal host has legitimate business reasons to access `.ru` domains.
- Reviewing associated traffic for other indicators of compromise (e.g., outbound connections, file execution).

User

What IP addresses were contacted during each kind of Priority 1 alert?

User

The following citations were found in the files provided by the user:

Citation 1: "[] **[120:8:2] (http_inspect) INVALID CONTENT-LENGTH OR CHUNK SIZE []**
[Classification: Unknown Traffic] [Priority: 3]
06/27-13:44:27.380600 192.168.1.96:49869 -> 79.170.44.118:80
TCP TTL:128 TOS:0x0 ID:8072 IpLen:20 DgmLen:40 DF
A*** Seq: 0xEFFD07E5 Ack: 0x73B800A1 Win: 0x100 TcpLen: 20
[Xref => http://cve.mitre.org/cgi-bin/cvename.cgi?name=2013-2028]

[] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:27.423181 192.168.1.96:49455 -> 104.28.7.63:80
TCP TTL:53 TOS:0x0 ID:17366 IpLen:20 DgmLen:1477 DF
A*** Seq: 0xB4960B24 Ack: 0x2701F719 Win: 0x8400 TcpLen: 20

[] [120:3:1] (http_inspect) NO CONTENT-LENGTH OR TRANSFER-ENCODING IN HTTP RESPONSE []
[Classification: Unknown Traffic] [Priority: 3]
06/27-13:44:27.380124 79.170.44.118:80 -> 192.168.1.96:49869
TCP TTL:45 TOS:0x0 ID:62263 IpLen:20 DgmLen:106 DF
AP Seq: 0x73B8005F Ack: 0xEFFD07E6 Win: 0x0 TcpLen: 20"

Citation 2: "[] **[120:3:1] (http_inspect) NO CONTENT-LENGTH OR TRANSFER-ENCODING IN HTTP RESPONSE []**
[Classification: Unknown Traffic] [Priority: 3]
06/27-13:44:33.298262 82.208.109.253:80 -> 192.168.1.96:49934

TCP TTL:42 TOS:0x0 ID:40173 IpLen:20 DgmLen:206 DF
AP Seq: 0xAD5DCD6C Ack: 0xF8205E2C Win: 0xFF00 TcpLen: 20

[] [120:3:1] (http_inspect) NO CONTENT-LENGTH OR TRANSFER-ENCODING IN HTTP RESPONSE []
[Classification: Unknown Traffic] [Priority: 3]
06/27-13:44:32.898226 82.208.109.253:80 -> 192.168.1.96:49930
TCP TTL:42 TOS:0x0 ID:27626 IpLen:20 DgmLen:206 DF
AP Seq: 0x41D33ACB Ack: 0x19706FE7 Win: 0xFF00 TcpLen: 20

[] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:28.798936 192.168.1.96:49892 -> 198.185.159.144:80
TCP TTL:128 TOS:0x0 ID:8272 IpLen:20 DgmLen:1571 DF
AP Seq: 0xE0EDD4EC Ack: 0x46F626D7 Win: 0x6DA8 TcpLen: 20"

Citation 3: "**[] [120:3:1] (http_inspect) NO CONTENT-LENGTH OR TRANSFER-ENCODING IN HTTP RESPONSE []**
[Classification: Unknown Traffic] [Priority: 3]
06/27-13:44:12.085410 192.154.109.132:80 -> 192.168.1.96:49312
TCP TTL:128 TOS:0x0 ID:3759 IpLen:20 DgmLen:16549 DF
A* Seq: 0x8631C313 Ack: 0x27471774 Win: 0x0 TcpLen: 20

[] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.108344 192.168.1.96:49325 -> 94.23.16.70:80
TCP TTL:105 TOS:0x8 ID:29917 IpLen:20 DgmLen:1581 DF
A* Seq: 0x6C93058F Ack: 0xFD05DC18 Win: 0x0 TcpLen: 20

[] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection []
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.118376 192.168.1.96:49337 -> 213.142.145.224:80
TCP TTL:108 TOS:0x0 ID:1943 IpLen:20 DgmLen:1245 DF
A* Seq: 0x4537192 Ack: 0x763103A2 Win: 0x0 TcpLen: 20"

Use the citations above to respond to the user query, only if they are relevant. Otherwise, respond to the best of your ability without them.

User Query:

What's the thing with the SMTP traffic? I can see many SMTP packets sent to the infected computer. Why is this happening and is it relevant to the coursework?

Assistant

The presence of SMTP traffic sent **to** the infected computer (`192.168.1.96`) is a classic indicator of a **phishing delivery mechanism**. Here's why:

Why This Happens

- **Phishing Email Delivery**: Attackers use compromised email servers (or legitimate SMTP relays) to send malicious emails containing links or attachments that deliver malware to the victim. The infected host is receiving these emails, which may contain web-delivered payloads (like JavaScript-based trojans).
- **Web Delivery via HTTP/S**: As seen in the logs, the malware appears to be delivered through web requests (port 80) to external IPs, possibly after being accessed via a phishing link.

Relevance to Coursework

This is highly relevant because:

1. It aligns with common malware distribution vectors (phishing → web delivery → C2).
2. The combination of SMTP traffic with obfuscated JavaScript and CNC trojan alerts suggests a multi-stage attack, which is a core topic in incident response coursework.

Recommended Grep Commands

```
# Extract all SMTP-related alerts involving the infected host
grep -i "smtp" CW1Snort.txt | grep "192.168.1.96"

# Check for phishing or malicious links associated with the email delivery
grep -iE "(phish|malicious|suspicious)" CW1Suricata.txt | grep "192.168.1.96"
```

User

The following citations were found in the files provided by the user:

Citation 1: "[] **[139:1:1] (spp_sdf) SDF Combination Alert** []
[Classification: Senstive Data] [Priority: 2]
06/27-13:43:59.242880 59.106.164.230 -> 192.168.1.96
PROTO:254 TTL:128 TOS:0x0 ID:1061 IpLen:20 DgmLen:20 DF

[] [120:8:2] (http_inspect) INVALID CONTENT-LENGTH OR CHUNK SIZE []
[Classification: Unknown Traffic] [Priority: 3]
06/27-13:44:01.462986 192.168.1.96:49200 -> 96.82.200.1:80
TCP TTL:128 TOS:0x0 ID:1284 IpLen:20 DgmLen:40 DF
A* Seq: 0x763F0FC7 Ack: 0xC5FE1D52 Win: 0xFB6D TcpLen: 20
[Xref => http://cve.mitre.org/cgi-bin/cvename.cgi?name=2013-2028]

[] [1:28629:6] INDICATOR-OBFUSCATION obfuscated script encoding detected []
[Classification: Misc activity] [Priority: 3]
06/27-13:44:01.624079 148.251.33.194:80 -> 192.168.1.96:49194
TCP TTL:43 TOS:0x0 ID:37796 IpLen:20 DgmLen:11720 DF
A* Seq: 0xA2F1C810 Ack: 0x95857EE1 Win: 0x7F TcpLen: 20"

Citation 2: "**[] [139:1:1] (spp_sdf) SDF Combination Alert []**
[Classification: Senstive Data] [Priority: 2]
06/27-13:38:34.536182 119.28.70.207 -> 192.168.1.96
PROTO:254 TTL:128 TOS:0x0 ID:661 IpLen:20 DgmLen:20 DF

[] [1:15306:22] FILE-EXECUTABLE Portable Executable binary file magic detected []
[Classification: Potential Corporate Privacy Violation] [Priority: 1]
06/27-13:43:52.407982 145.131.10.21:80 -> 192.168.1.96:49190
TCP TTL:239 TOS:0x20 ID:41885 IpLen:20 DgmLen:1207
AP Seq: 0xA90C366B Ack: 0xCB5AB7FC Win: 0x2180 TcpLen: 20

[] [1:15306:22] FILE-EXECUTABLE Portable Executable binary file magic detected []
[Classification: Potential Corporate Privacy Violation] [Priority: 1]
06/27-13:43:52.407982 145.131.10.21:80 -> 192.168.1.96:49190
TCP TTL:239 TOS:0x20 ID:41885 IpLen:20 DgmLen:1207
AP Seq: 0xA90C366B Ack: 0xCB5AB7FC Win: 0x2180 TcpLen: 20"

Citation 3: "**[] [124:1:1] (smtp) Attempted command buffer overflow: more than 512 chars []**
[Classification: Attempted Administrator Privilege Gain] [Priority: 1]
06/27-13:44:39.007762 192.168.1.96:50074 -> 88.208.248.90:25
TCP TTL:128 TOS:0x0 ID:9645 IpLen:20 DgmLen:661 DF
AP Seq: 0x639AB485 Ack: 0x958D7E10 Win: 0x3E00 TcpLen: 20
[Xref => http://www.microsoft.com/technet/security/bulletin/ms05-021.mspx][Xref (http://www.microsoft.com/technet/security/bulletin/ms05-021.mspx%5D%5BXref) => http://cve.mitre.org/cgi-bin/cvename.cgi?name=2005-0560][Xref (http://cve.mitre.org/cgi-bin/cvename.cgi?name=2005-0560%5D%5BXref) => http://cve.mitre.org/cgi-bin/cvename.cgi?name=2001-0260]

[] [120:3:1] (http_inspect) NO CONTENT-LENGTH OR TRANSFER-ENCODING IN HTTP RESPONSE []
[Classification: Unknown Traffic] [Priority: 3]
06/27-13:44:33.711965 82.208.109.253:80 -> 192.168.1.96:49969
TCP TTL:42 TOS:0x0 ID:1308 IpLen:20 DgmLen:206 DF
AP Seq: 0x871854F6 Ack: 0xEC0FC493 Win: 0xFF00 TcpLen: 20"

Use the citations above to respond to the user query, only if they are relevant. Otherwise, respond to the best of your ability without them.

User Query:

Write an example of concise sentence on this SMTP evidence for the report

Assistant

The SMTP evidence shows `192.168.1.96` attempted a command buffer overflow (>512 chars) to external IP `88.208.248.90:25`, referencing CPEs 2001-0260, 2005-0560, and MS05-021 (Mail Exchange Server vulnerability), indicating an exploit attempt targeting the host's mail server for privilege escalation.